

Learning target 1.6, version 1

Explain the relationships between a function, its derivative, and its second derivative. Interpret information about these objects in real-world contexts.

Students in a recent calculus class were playing a Mad Libs game about the relationships between shapes of functions and values of the derivative:

Whenever the _____ is _____, the _____ is _____.
 noun adjective noun adjective

For each of the sentences they generated below, explain why it is correct or incorrect.
(Hint: I don't think it's possible to generate a good explanation without saying the word "slope.")

- (a) Whenever the original function is decreasing, the derivative is negative.
- (b) Whenever the derivative is increasing, the original function is positive.
- (c) Whenever the second derivative is positive, the original function is concave up.
- (d) Whenever the derivative is increasing, the second derivative is positive.